Recovering a $\mathbb{Q}(q,t)$-basis from specializations: what the reconstruction proves

We reconstruct the Gröbner basis of a parametric ideal over $K = \mathbb{Q}(q,t)$ by solving the ideal at many integer points (q_0, t_0) and interpolating. This note states what that procedure does and does not prove. The two theorems are standard; the point is to separate the *search* (interpolation, heuristic) from the *proof* (a check with q, t transcendental), and to say precisely where the objection "a discrete grid may sit on the bad locus" bites.

1 Setup

Let $\Bbbk = \mathbb{Q}$, $R = \Bbbk[q,t]$, $K = \Bbbk(q,t)$. Fix an alphabet $X = \{x_1, \dots, x_{n-1}, y_1, \dots, y_{n-1}\}$ and an admissible well-order $\prec$ on the free monoid $\langle X\rangle$ (compatible with two-sided multiplication). Let $f_1, \dots, f_m \in R\langle X\rangle$ be the presentation (braid, inverse, zipper and untwisting relations; coefficients in $\mathbb{Z}[q,t]$), $I = {}_K(f_1, \dots, f_m)_K$ the two-sided ideal they generate in $K\langle X\rangle$, and $\mathcal{A} = K\langle X\rangle/I$. We assume $\dim_K \mathcal{A} < \infty$, which holds in our case.

For a point $\mathfrak{p} = (q_0, t_0) \in \mathbb{A}^2_{\Bbbk}(\Bbbk) = \Bbbk^2$ let $\sigma_{\mathfrak{p}} \colon R\langle X\rangle \to \Bbbk\langle X\rangle$ be the specialization $q \mapsto q_0,\ t \mapsto t_0$, put $I_{\mathfrak{p}} = {}_{\Bbbk}(\sigma_{\mathfrak{p}} f_1, \dots, \sigma_{\mathfrak{p}} f_m)$ and $\mathcal{A}_{\mathfrak{p}} = \Bbbk\langle X\rangle/I_{\mathfrak{p}}$.

For an ideal J (over K or $\Bbbk$), noncommutative Buchberger–Mora reduction w.r.t. $\prec$ terminates when J is co-finite, producing the unique reduced Gröbner basis $G(J)$ (monic leading words, tails supported on standard monomials). Write $G = G(I) \subseteq K\langle X\rangle$ and $G_{\mathfrak{p}} = G(I_{\mathfrak{p}}) \subseteq \Bbbk\langle X\rangle$. The set of standard monomials $\mathrm{Std}(J) = \{\, w \in \langle X\rangle : \mathrm{lm}(g) \nmid w \ \forall g \in G(J) \,\}$ is a basis of the quotient. Each $g \in G$ has the form

$$g \;=\; m_g \;+\; \sum_{w \in T_g} c_{g,w}\, w, \qquad c_{g,w} \in K, \tag{1}$$

with $m_g = \mathrm{lm}(g)$ and $T_g \subset \mathrm{Std}(I)$ finite. *Reconstruction* means: recover the coefficient functions $c_{g,w} \in K$ from their values $c_{g,w}(\mathfrak{p})$ at grid points.

2 Specialization commutes with the Gröbner basis off a hypersurface

Theorem 1. *There is a nonzero $d \in R$ such that for every $\mathfrak{p}$ with $d(\mathfrak{p}) \neq 0$:*

(i) *every $c_{g,w}$ lies in the localization $R_d = R[d^{-1}]$, so $\sigma_{\mathfrak{p}} g$ is defined;*

(ii) *$\sigma_{\mathfrak{p}} G := \{\sigma_{\mathfrak{p}} g : g \in G\}$ is the reduced Gröbner basis of $I_{\mathfrak{p}}$;*

(iii) *$\mathrm{lm}(\sigma_{\mathfrak{p}} g) = m_g$, $\mathrm{Std}(I_{\mathfrak{p}}) = \mathrm{Std}(I)$, and $\dim_{\Bbbk} \mathcal{A}_{\mathfrak{p}} = \dim_K \mathcal{A}$.*

Proof. Run Buchberger–Mora on $\{f_i\}$ over K; it terminates, giving G, using finitely many operations of K. Let $d \in R$ be a common denominator of all $c_{g,w}$ and of every leading coefficient that was inverted (used as a pivot) during the run. Then $G \subseteq R_d\langle X\rangle$, which is (i).

Two families of witnesses are produced by (or extractable from) the run, each a *finite* identity with coefficients in R_d:

$$\textbf{(R)}\quad f_i = \sum_j p_{ij}\, g_{k_j}\, q_{ij}\ \ (f_i \text{ reduces to } 0 \bmod G), \qquad \textbf{(S)}\quad S(g,g') = \sum_j p'_j\, g_{k'_j}\, q'_j$$

for every f_i and every overlap ambiguity $S(g,g')$ of G; here p, q are words. Because $d(\mathfrak{p}) \neq 0$, $\sigma_\mathfrak{p}$ restricts to a ring homomorphism $R_d \to \Bbbk$, so it carries (R) and (S) to the same identities over $\Bbbk$ for $\sigma_\mathfrak{p}G$. Thus every $\sigma_\mathfrak{p}f_i$ reduces to 0 modulo $\sigma_\mathfrak{p}G$, and every S-polynomial of $\sigma_\mathfrak{p}G$ reduces to 0 modulo $\sigma_\mathfrak{p}G$. The leading words are unchanged (the m_g have coefficient 1), so Buchberger's criterion applies: $\sigma_\mathfrak{p}G$ is a Gröbner basis of $(\sigma_\mathfrak{p}G)$. From (R), $I_\mathfrak{p} \subseteq (\sigma_\mathfrak{p}G)$; from a cofactor expression $g = \sum p\, f_i\, q$ over R_d (each $g \in I$), $\sigma_\mathfrak{p}g \in I_\mathfrak{p}$, so $(\sigma_\mathfrak{p}G) \subseteq I_\mathfrak{p}$. Hence $(\sigma_\mathfrak{p}G) = I_\mathfrak{p}$ and $\sigma_\mathfrak{p}G$ is a Gröbner basis of $I_\mathfrak{p}$ with leading words $\{m_g\}$; this gives $\mathrm{Std}(I_\mathfrak{p}) = \mathrm{Std}(I)$ and the equal dimensions. Monic leading terms and standard tails are preserved by $\sigma_\mathfrak{p}$, so $\sigma_\mathfrak{p}G$ is reduced; uniqueness of the reduced Gröbner basis gives $\sigma_\mathfrak{p}G = G_\mathfrak{p}$. This is (ii), (iii). □

Write $U = D(d) = \{\, d \neq 0 \,\}$ (a dense open subset of $\mathbb{A}^2$) and $B = V(d) = \mathbb{A}^2 \setminus U$.

Corollary 2. *For $\mathfrak{p} \in U$, the reduced Gröbner basis returned at $\mathfrak{p}$ carries, in slot (g,w) of (1), the value $c_{g,w}(\mathfrak{p})$. On U the numbers we compute are point-values of the generic coefficient functions; recovering $c_{g,w}$ from such values is interpolation.*

3 Why sampling alone cannot prove the recovery

$B = V(d)$ is a proper Zariski-closed subset, hence a curve (a nonzero d cuts out something of codimension ≥ 1). This does *not* make $B(\Bbbk)$ finite: any component that is a rational curve — e.g. a line $\{q = 1\}$ or $\{t + q^2 = 0\}$ — carries infinitely many rational points.

Proposition 3. *No function of the samples $\{G_\mathfrak{p} : \mathfrak{p} \in \Gamma\}$ on a finite grid $\Gamma \subset \Bbbk^2$ certifies $\Gamma \subseteq U$. In particular, constancy of $\dim_\Bbbk \mathcal{A}_\mathfrak{p}$ across Γ does not.*

Proof. The map $\mathfrak{p} \mapsto \dim_\Bbbk \mathcal{A}_\mathfrak{p} = \#\,\mathrm{Std}(I_\mathfrak{p})$ is upper semicontinuous: for a co-finite family it satisfies $\dim_\Bbbk \mathcal{A}_\mathfrak{p} \geq \dim_K \mathcal{A}$ with equality exactly on a dense open set containing U (Theorem 1(iii); semicontinuity of fibre dimension). Its level sets are locally closed, and a level set on which the value exceeds $\dim_K \mathcal{A}$ can be a positive-dimensional subvariety of B. A grid Γ contained in such a level set is consistent (constant dimension, constant leading-word set) yet lies entirely in B, and then the interpolant of Corollary 2 reconstructs the coefficient functions of the *restricted* family, not of G. Nothing intrinsic to $\{G_\mathfrak{p} : \mathfrak{p} \in \Gamma\}$ distinguishes this case. □

Thus interpolation is a heuristic search: it returns a candidate $\widetilde{G}$, correct if $\Gamma \subseteq U$, but the samples cannot warrant $\Gamma \subseteq U$.

4 What proves the recovery: a check with q, t transcendental

The proof is decoupled from the sampling. It runs over K, so the bad locus is irrelevant.

Theorem 4. *Let $\widetilde{G} = \{\tilde{g}_1, \ldots, \tilde{g}_N\} \subset K\langle X\rangle$ be finite, with monic, pairwise-nondividing leading words $m_\ell = \mathrm{lm}(\tilde{g}_\ell)$ and tails supported on $\{w : w \prec m_\ell,\ m_{\ell'} \nmid w\ \forall \ell'\}$. Suppose, over K:*

(a) *each f_i reduces to 0 modulo $\widetilde{G}$;*

(b) *every overlap S-polynomial of $\widetilde{G}$ reduces to 0 modulo $\widetilde{G}$;*

(c) *each $\tilde{g}_\ell \in I$, exhibited by a two-sided cofactor $\tilde{g}_\ell = \sum_j p_j\, f_{i_j}\, q_j$.*

Then $\widetilde{G} = G$.

Proof. By (b) and Bergman's diamond lemma / Buchberger's criterion, $\widetilde{G}$ is a Gröbner basis of $J := (\widetilde{G})$. By (c), $J \subseteq I$; by (a), $I \subseteq J$. Hence $J = I$ and $\widetilde{G}$ is a Gröbner basis of I; it is monic with standard tails, hence reduced; by uniqueness $\widetilde{G} = G$. □

Conditions (a)–(c) contain no specialization: they are identities in $K\langle X\rangle$ with q, t transcendental, so Proposition 3 does not apply to them. They are finite (normal-form computations and cofactor expansions). Interpolation supplies the candidate $\widetilde{G}$; Theorem 4 certifies it. This is the whole logic: *search by specialization, prove by transcendence.*

Remark 5 (verifying (a)–(c) by evaluation, without symbolic algebra)**.** Each reduction in (a),(b) and each cofactor in (c) is, after clearing the common denominator d, a polynomial identity $P = 0$ in $R\langle X\rangle$: word by word, a coefficient identity in $R = \mathbb{Z}[q, t]$ of some bidegree (D_q, D_t) *read off the cleared certificate.* A polynomial in R of bidegree (D_q, D_t) vanishing on a $(D_q{+}1) \times (D_t{+}1)$ integer product grid is identically zero (iterated univariate interpolation / finite differences). So (a)–(c) are verified by exact evaluation on such a grid — deterministic, no probability, no genericity. This is again a use of points, but now the points check an identity of *known* degree, so the finite grid *does* suffice; contrast Proposition 3, where the unknown was a rational function of unbounded a priori degree tied to the unknown locus B.

5 Atoms

By construction d is a product of pivots, each a nonzero element of R. Let $a_1, \ldots, a_r$ be the distinct irreducible factors of d in R — the *atoms*.

Proposition 6.

(i) *Every coefficient satisfies $c_{g,w} \in R[a_1^{-1}, \ldots, a_r^{-1}]$, i.e. $c_{g,w} = N_{g,w}/\prod_i a_i^{e_i}$ with $N_{g,w} \in R$. Clearing $\prod_i a_i^{e_i}$ turns the recovery of $c_{g,w}$ into polynomial interpolation of $N_{g,w}$.*

(ii) *$B = V(d) = \bigcup_i V(a_i)$, and the locus where $\dim_{\mathbb{k}} \mathcal{A}_\mathfrak{p} > \dim_K \mathcal{A}$ is contained in B. The atoms index the components of the degeneration locus.*

(iii) *Any K-valued invariant built from $\mathcal{A}$ by the same field operations lies in $R[d^{-1}]$ after enlarging d by finitely many further pivots; e.g. the Gram determinant $\det\Theta$ of the regular trace form satisfies $\det\Theta \in R[d^{-1}]$, so for $\mathfrak{p} \in U$ the algebra $\mathcal{A}_\mathfrak{p}$ is semisimple iff $\det\Theta(\mathfrak{p}) \neq 0$, and the non-semisimple locus in U is cut out by the extra irreducible factors of $\det\Theta$ — again atoms.*

This is the single mechanism behind the recurring factors: the atoms are the irreducible components of the discriminantal hypersurface B carrying *all* denominators produced by the same elimination — the Gröbner tails, the X_3-cofactor denominator $t^4(t + q^2)$, and the semisimplicity locus.

6 A rigorous sampling route, and why we do not use it

Proposition 3 is an obstruction only because $\deg d$ is unknown. Grant an a priori bound $\deg d \le \Delta$ and a numerator bound $\deg N_{g,w} \le \delta$. A component $V(a_i)$ meets a product grid $\Gamma = Q \times T$ in at most $(\deg a_i)\,|T|$ points (a plane curve of degree e meets $|T|$ vertical lines in $\le e|T|$ points), so $|\Gamma \cap B| \le \Delta\,|T|$. A rational recovery of Berlekamp–Welch type over Γ then returns the true $N_{g,w}/\prod a_i^{e_i}$ and locates the bad points, provided $|\Gamma|$ exceeds the interpolation requirement plus twice the error bound. This *is* a proof from samples. It fails in practice only because the general bounds on Δ (through the sizes and coefficient degrees of the Buchberger matrices) are far too large to make $|\Gamma|$ feasible. Hence we certify a posteriori by Theorem 4 rather than a priori by error correction: the candidate is cheap to guess and cheap to verify, whereas the guaranteed-correct interpolation would need an infeasible grid.

Summary

Theorem 1 identifies what the samples are: on $U = D(d)$ they are point-values of the generic Gröbner coefficients, and recovery is interpolation. Proposition 3 is the honest limitation: a finite grid cannot certify $\Gamma \subseteq U$, so interpolation alone proves nothing. Theorem 4 supplies the proof, over K with q, t transcendental, where the bad locus has no meaning; Remark 5 makes that check a finite exact evaluation. Proposition 6 explains the denominators and ties them to the degeneration and semisimplicity loci. Reconstruction is search by specialization, proof by transcendence.

Notes and references

That a Gröbner basis over the rational-function field in the parameters specializes to a Gröbner basis of the specialized ideal, provided a distinguished polynomial does not vanish at the specialization point, is due to Gianni [3] and, in the leading-term-stability form used in Theorem 1, to Kalkbrener [2]. The systematic parametric theory is Weispfenning's comprehensive Gröbner bases [1] and the canonical Gröbner cover of Montes and Wibmer [4], which partitions parameter space into locally closed strata carrying fixed reduced-basis data; the good locus U of Theorem 1 is the open stratum, and B its complement. Becker and Weispfenning [5] is the standard text.

The algebras here are quotients of a free associative algebra, so "Gröbner basis" is meant in the two-sided noncommutative sense. Uniqueness of the reduced basis and the confluence criterion by overlap ("ambiguity") S-polynomials, used in Theorems 1 and 4, are Bergman's diamond lemma [6] and Mora's noncommutative Buchberger procedure [7, 8]; see also Green [9] and Ufnarovski [10].

Upper semicontinuity of the fibre dimension in Proposition 3 is standard: Hartshorne [11] (semicontinuity of fibre dimension), Eisenbud [12] (flatness and the Hilbert function), Matsumura [13].

Recovery of a coefficient from its values (Corollary 2, Remark 5) is multivariate interpolation: dense and sparse polynomial interpolation [14, 15], black-box separation of numerator and denominator [16], multivariate rational interpolation [17], and classical Thiele/reciprocal-difference rational interpolation [18, §2.2]. The exact-$\mathbb{Q}$ value at each grid point is itself reconstructed from modular images — the inner layer — by modular and traced Gröbner computation [23, 24, 25] with rational-number reconstruction [26].

The error-correcting recovery of §7 is the decoding view of interpolation: the Welch–Berlekamp algorithm [19, 20], its list-decoding extension [21], and sparse-interpolation codes decoding beyond half the minimum distance [22].

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